

$$1. a. (315, 84) \rightarrow 315/84 = 3 + 63$$

$$84/63 = 1 + 21$$

$$\text{gcd} = 21$$

$$63/21 = 3$$

$$b. 21 = 315m + 84n$$

$$21 = 84 - 63(1)$$

$$21 = 84 - 315 + 252$$

$$21 = -315 + 84 + 252$$

$$-315 + 336$$

$$(-1) \cdot 315 + 4 \cdot 84$$

$$\therefore m = -1$$

$$n = 4$$

$$2. a) \text{gcd}(a, b) = \text{gcd}(a + kb, b) \text{ for } a \geq b \geq 0 \checkmark$$

$$b) \text{Counter example } a = 6, b = 4$$

$$\text{gcd}(6, 4) = 2$$

$$\text{gcd}(36, 4) = 4$$

$\therefore$ Does not hold

$$3. 1 \times \text{last } x \cdot 5 + y$$

$$\text{where } n = 7k = 10a + b$$

$$n = \overset{35}{7}k = \overset{50}{10}a + \overset{5b}{b}$$

$$\rightarrow 35k = 50a + 5b$$

$$35k - 50a = 5b + a$$

$$7(5k - 7a) = 5b + a$$

Assume $a + 5b$ is a multiple of 7

Let n be an integer $n = 10a + b$ a = the number except the last digit and b is the last digit
 $0 \leq b < 9$

If $a + 5b$ is a multiple of 7, then n is divisible by 7.

$$5n = 50a + 5b = 49a + (a + 5b)$$

as 49 is divisible by 7, $5n = a + 5b \pmod{7}$